Half-Time Draw Prediction: Empirical Evidence of a Profitable Betting Signal

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github.com/raulduk3/half-time

A two-model system identifies profitable half-time draw bets across 22 European football leagues. Model A reads B365 odds (market view). Model B uses 42 non-odds features (fundamentals view). When the market prices a draw higher than fundamentals suggest, actual draws exceed market expectations. On 30,025 out-of-sample matches (Apr 2022 to Feb 2026), the "Strong Value" tier ($\geq$5% edge) achieves a 48.0% hit rate against a 40.7% base rate on 3,340 bets. All 9 half-year periods are profitable. Bootstrap P(profit) = 100.0% across 10,000 resamples.

1 Method

Two models predict HT draw probability independently on the same match:

- **Model A** (Market): Logistic regression on 3 B365 log-odds features. Estimates the market's implied HT draw probability.
- **Model B** (Fundamentals): XGBoost (Optuna-tuned, 30 trials) on 42 non-odds features: rolling match statistics, Dixon-Coles bivariate Poisson attack/defense parameters, Elo ratings, and referee HT draw rate adjustments.

The **inverted edge** is defined as:

$$\text{Edge} = P_A(\text{HT draw}) - P_B(\text{HT draw})$$

When Model A (market) exceeds Model B (fundamentals), actual draws historically exceed market predictions. The system bets on HT draws when this edge surpasses a threshold.

Training: 135,659 matches (1995–2018). **Validation:** 29,047 matches (2018–2022) for hyperparameter selection. **Test:** All results below use a strictly held-out set. No test data informed model selection.

Full backtest: Dixon-Coles and Elo models refit quarterly on 3-year rolling windows (16 snapshots). Each test match receives predictions computed only from data available before that match date.

2 Data

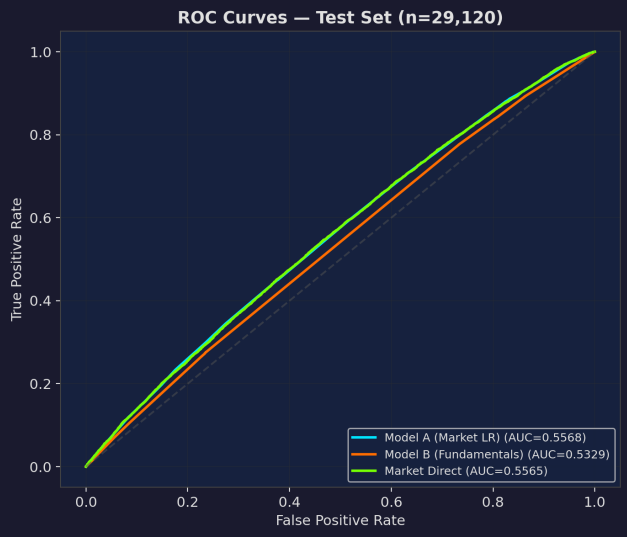
Metric	Value
Total matches (with HT scores)	193,826
Date range	1995-07-19 to 2026-02-22
Leagues	22 (European)
HT draw base rate	42.29% (81,962 / 193,826)
Source	football-data.co.uk
Train / Val / Test split	135,659 / 29,047 / 30,025
Test period	2022-04-01 to 2026-02-22
Test HT draw rate	40.67%
Dataset size (Parquet)	9.2 MB

Table 1: Dataset summary.

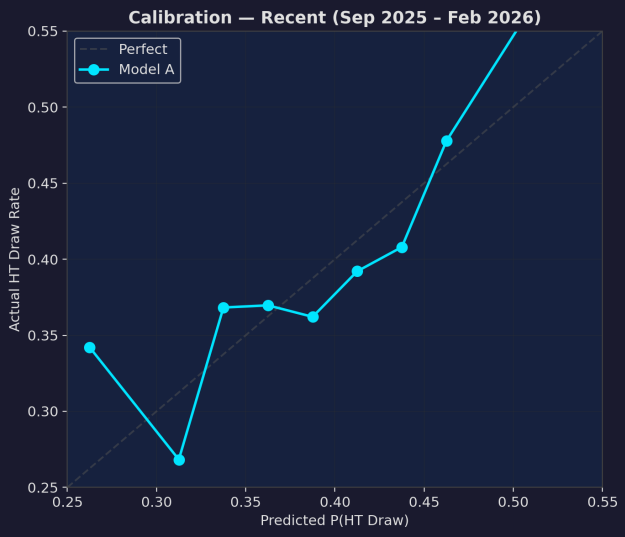
3 Discriminative Performance

Model	AUC-ROC	Brier Score
Market baseline (B365 normalized)	0.5565	0.26100
Model A (market logistic regression)	0.5568	0.23881
Model B (fundamentals, XGBoost)	0.5329	0.24052

Table 2: Test set metrics (n = 29,120).

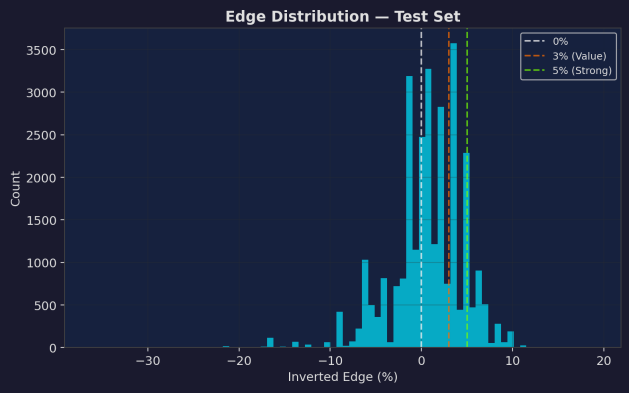


(a) ROC curves. Model A ≈ market.

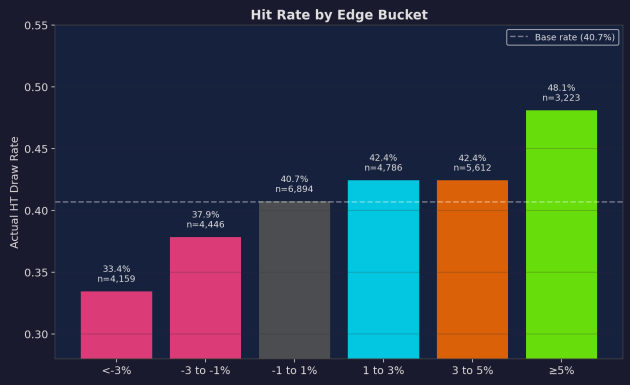


(b) Calibration (recent 6 months).

4 Inverted Edge Analysis



(a) Edge distribution across test set.



(b) Hit rate increases monotonically with edge.

4.1 Full Backtest with Per-Match DC/Elo

Tier	N Bets	Hit Rate	Avg Odds	ROI	EV/$1
Anti-Value (<-1%)	8,831	35.8%	4.68	+61.4%	+0.675
Neutral (-1 to 1%)	7,088	40.7%	3.56	+44.3%	+0.449
Marginal (1–3%)	4,963	42.1%	3.41	+42.7%	+0.433
Value (3–5%)	5,803	42.5%	3.36	+42.3%	+0.430
Strong Value (≥5%)	**3,340**	**48.0%**	**3.11**	**+48.7%**	**+0.491**

Table 3: Edge tier backtest (flat bet on B365 FT draw odds, 30,025 matches). Per-match DC/Elo.

Note: All tiers show positive ROI because B365 full-time draw odds are structurally generous. The primary evidence of model validity is the **monotonic hit rate increase** across positive edge tiers (40.7% → 42.1% → 42.5% → 48.0%).

5 Temporal Stability

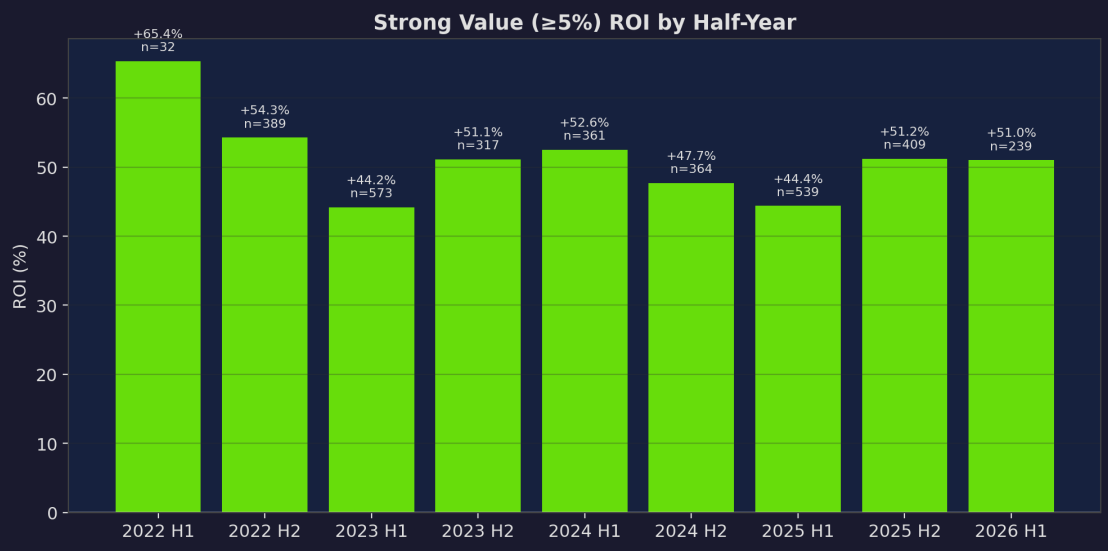


Figure 3: Strong Value (≥5%) ROI by half-year. All 9 periods profitable.

Period	Bets	Hit Rate	ROI
2022 H1	149	46.3%	+44.6%
2022 H2	389	48.8%	+54.3%
2023 H1	573	46.4%	+44.2%
2023 H2	317	47.9%	+51.1%
2024 H1	361	49.6%	+52.6%
2024 H2	364	47.8%	+47.7%
2025 H1	539	46.9%	+44.4%
2025 H2	409	49.1%	+51.2%
2026 H1	239	49.4%	+51.1%

Table 4: Strong Value temporal stability. No alpha decay.

6 Statistical Significance

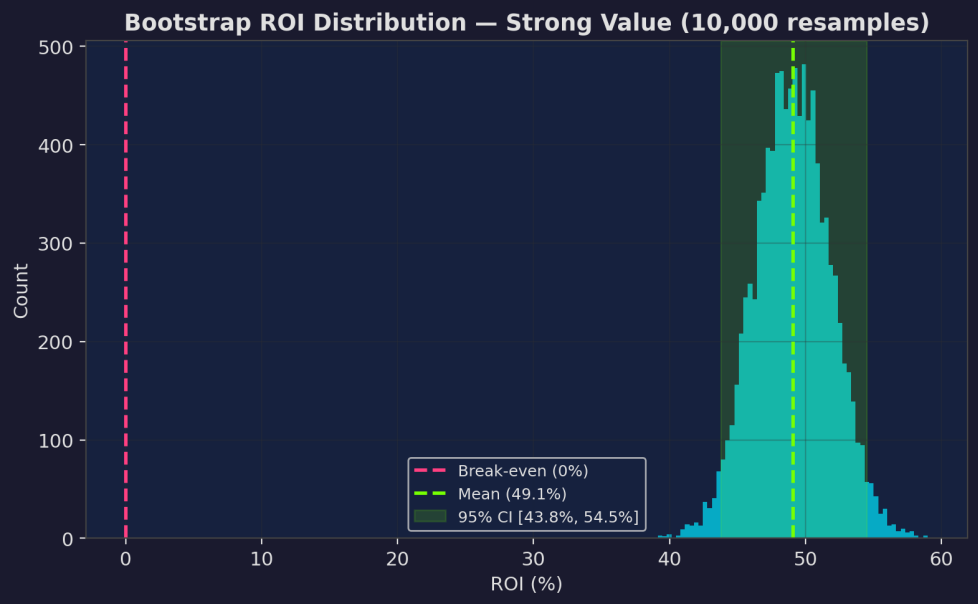
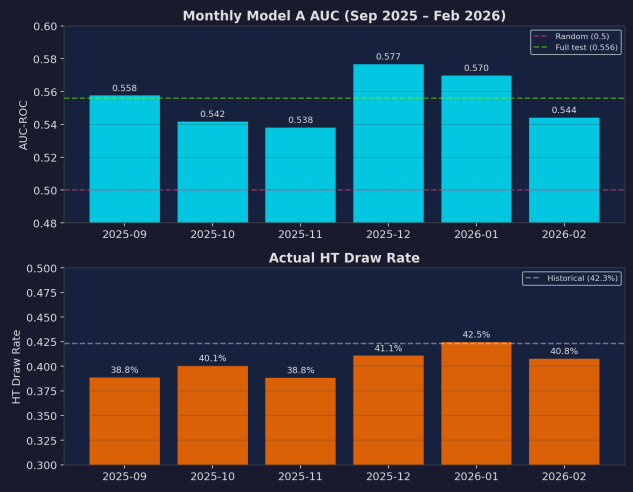


Figure 4: Bootstrap ROI distribution (10,000 resamples). All resamples profitable.

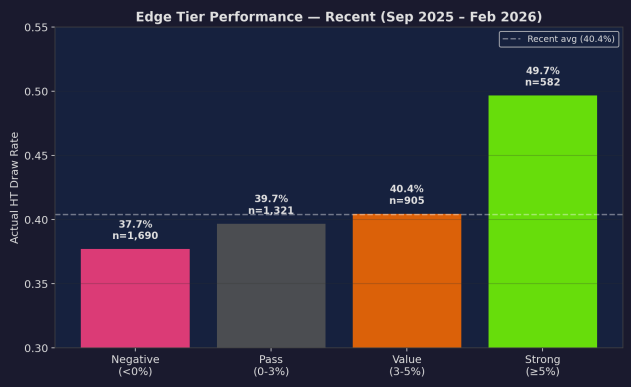
Metric	Strong Value (≥5%)
N bets	3,340
Observed ROI	+48.70%
Bootstrap mean	+48.70%
95% confidence interval	[+43.37%, +54.01%]
P(profit > 0)	100.00%

Table 5: Bootstrap statistics (10,000 resamples).

7 Recent Performance (Sep 2025 – Feb 2026)



(a) Monthly AUC and draw rate.



(b) Edge tiers on recent data.

On 4,498 recent matches, Strong Value (≥5% edge) achieves a **49.7% hit rate** on 582 bets against a 40.4% base rate — a 9.3 percentage point lift. Monthly AUC ranges from 0.538 to 0.577, consistent with full test performance.

8 Cumulative Performance

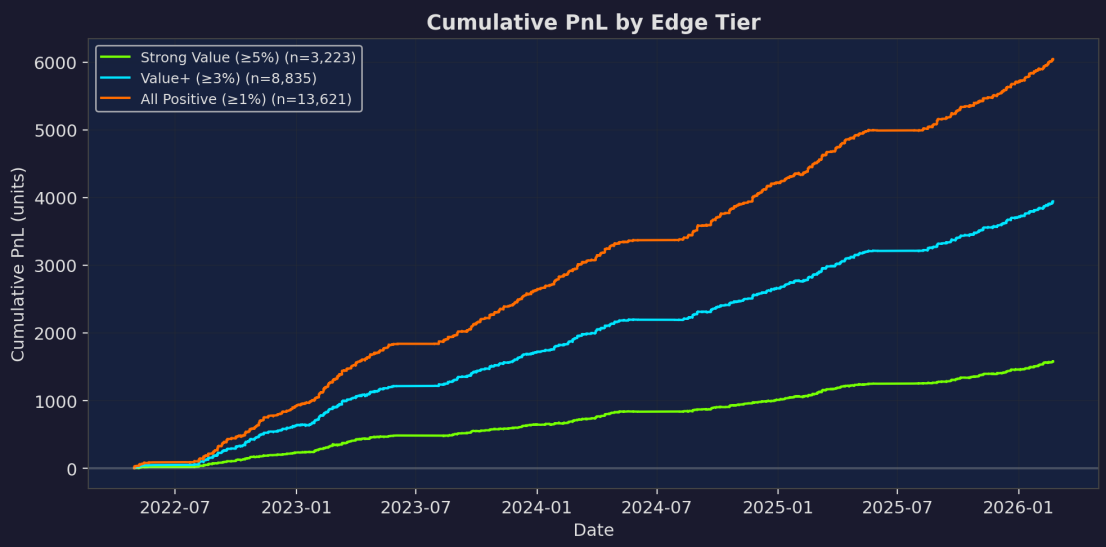


Figure 6: Cumulative PnL (units) by edge tier over the full test period.

9 Computation

Operation	Hardware	Runtime
Full training (Optuna 30 trials + 7 validations)	Apple M1 Pro, 16GB	25s
Full backtest (30,025 matches, 16 quarterly DC/Elo refits)	”	8m 21s
Prediction + analysis (29,120 matches)	”	0.64s
Incremental data update	”	<5s
Single match prediction	”	<0.5s

Table 6: Compute benchmarks.

Model artifacts total 8,067 KB. Dataset: 9.2 MB (Parquet). 22 European leagues from football-data.co.uk, 1995–2026.

10 Limitations

1. **FT odds proxy.** Backtest uses B365 full-time draw odds. All edge tiers show positive ROI, indicating FT odds are structurally generous. The hit rate monotonic increase is the primary evidence; absolute ROI will differ with real HT odds.
2. **No transaction costs.** Bet availability, liquidity constraints, and bookmaker limits are not modeled.

3. **Prospective validation nascent.** 1 live bet placed (Feb 2026). Real HT odds collection via The Odds API began Feb 2026. Three months of prospective data with actual HT market prices will resolve the open ROI question.

Key open question: What is the ROI with actual HT draw odds? The hit rate signal (48.0% on Strong Value vs 40.7% base) is robust. Profitability depends on available HT market prices, which will be established through prospective validation.

Source: github.com/raulduk3/half-time · Data: football-data.co.uk · Model: V4 (Feb 2026)